

Automated Deep Learning–based Detection and Segmentation of Lung Tumors at CT Imaging

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See also the editorial by Langer in this issue.

Radiology 2025; 314(1):e233029 • <https://doi.org/10.1148/radiol.233029> • Content codes: **CH** **CT**

Background: Detection and segmentation of lung tumors on CT scans is critical for monitoring cancer progression, evaluating treatment responses, and planning radiation therapy; however, manual delineation is labor-intensive and subject to physician variability.

Purpose: To develop and evaluate an ensemble deep learning model for automating identification and segmentation of lung tumors on CT scans.

Materials and Methods: A retrospective study was conducted between July 2019 and November 2024 using a large dataset of CT simulation scans and clinical lung tumor segmentations from radiation therapy plans. This dataset was used to train a three-dimensional U-Net multiresolution ensemble model to detect and segment lung tumors on CT scans. Model performance was evaluated on internal and external test sets composed of CT simulation scans and lung tumor segmentations from two affiliated medical centers; scans included a single primary or metastatic lung tumor. Performance metrics included sensitivity, specificity, false-positive rate, and Dice similarity coefficient (DSC). Model-predicted tumor volumes were compared with physician-delineated volumes. Segmentation outcomes were compared using the Wilcoxon signed rank test or one-way analysis of variance, with $P < .05$ indicating a statistically significant difference.

Results: The model, trained on 1504 CT scans with clinical lung tumor segmentations, achieved 92% sensitivity (92 of 100) and 82% specificity (41 of 50) in detecting lung tumors on the combined test set of 150 CT scans. For tumor segmentations on the subset of 100 CT scans with a lung tumor, median model-physician DSC was 0.77 (IQR, 0.65–0.83), and median interphysician DSC was 0.80 (IQR, 0.72–0.86). Segmentation time was shorter for the model than for physicians (mean, 76.6 seconds vs 166.1 and 187.7 seconds; $P < .001$ for both).

Conclusion: Routinely collected radiation therapy data were useful for model training. The key strengths of the model include a three-dimensional U-Net ensemble approach for balancing volumetric context with resolution, robust tumor detection and segmentation performance, and generalizability to an external site.

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Supplemental material is available for this article.

Accurate detection and segmentation of lung tumors on medical images is important for various clinical applications, including defining radiation therapy targets, measuring tumor burden, and evaluating treatment response. Currently, experienced clinicians manually identify and segment lung tumors on medical images, which is labor-intensive and prone to inter- and intraphysician variability (1–5). Automated tumor delineation could improve the efficiency and consistency of tumor segmentation.

Deep learning methods, particularly convolutional neural networks, can learn hierarchical feature representations from a variety of input data and have been applied to various medical image analysis tasks, including tumor segmentation (6–12). Despite the large number of existing studies, many are limited by small training datasets, insufficient evaluation focused primarily on peripheral tumors, and model reliance on human input (eg, seed points) (13–17). Furthermore, most previous studies have focused on segmenting single primary lung tumors, while the segmentation of metastatic disease with multiple small lesions, a task well suited for automation, remains unexplored.

In this study, a large dataset of pre–radiation treatment CT simulation scans and associated clinical three-dimensional (3D) segmentations used for radiation therapy planning was used to develop a near-expert-level lung tumor detection and segmentation model. The primary aim was to develop a model that accurately identified and segmented lung tumors on CT scans from different medical centers.

Materials and Methods

This retrospective study was conducted between July 2019 and November 2024 and was approved by the Stanford University institutional review board.

Datasets

Four datasets of routinely collected pretreatment CT simulation scans and clinical tumor segmentations for patients undergoing radiation therapy at two affiliated medical centers were compiled (Fig 1). The training set comprised 1504 CT scans and the corresponding clinical lung tumor segmentations from 1295 patients treated for primary or metastatic lung tumors.

Abbreviations

DSC = Dice similarity coefficient, 3D = three dimensional

Summary

Using routine pre–radiation treatment CT simulation scans and tumor segmentation data, a deep learning model was developed to detect and segment lung tumors, and good performance was achieved on heterogeneous datasets.

Key Results

- In this retrospective study, a three-dimensional U-Net ensemble model was trained for lung tumor detection and segmentation using 1504 CT scans (1828 segmented lung tumors) and tested on 150 CT scans.
- The model achieved 92% sensitivity and 82% specificity in detecting lung tumors on 100 and 50 CT scans with and without lung tumors, respectively, and misidentified 0.78 tumors per scan.
- The median model-physician and physician-physician segmentation Dice similarity coefficients were 0.77 and 0.80, respectively.

Each scan in this dataset contained one or more lung tumors; however, not all tumors were consistently segmented (eg, some were excluded from the planned treatment and thus were not segmented). The internal test set consisted of 41 CT scans and clinical lung tumor segmentations for patients who underwent radiation therapy for primary lung tumors. The external test set

included 59 CT scans and clinical lung tumor segmentations for patients treated for primary or metastatic lung tumors at an affiliated medical center with different scanner equipment and procedures. Importantly, none of the patients in the test sets were in the training set. All CT scans in both the internal and external test sets contained only one lung tumor. Additionally, a non-lung tumor set of CT simulation scans derived from patients undergoing radiation therapy for breast cancer, none of which contained any lung tumors, was assembled. This set served as a negative control to assess the false-positive detection of lung tumors. The standard procedure for obtaining CT simulation scans at both medical centers was with breath hold; however, since the actual use of breath hold was not consistently recorded, detailed statistics on breath-hold versus free-breathing scans are unavailable.

Assessment of Clinical Tumor Segmentation Quality

A custom web application was designed to assess clinical tumor segmentation quality and clean the training set, as it consisted of routine clinical data and contained errors such as mislabeled or misaligned structures. The application displayed CT images and the corresponding segmentations; a radiation oncologist (M.F.G., 14 years or experience) reviewed these segmentations, labeling them as lung tumors, non-tumor structures, or misaligned contours. An example of this process is shown in Figure S1.

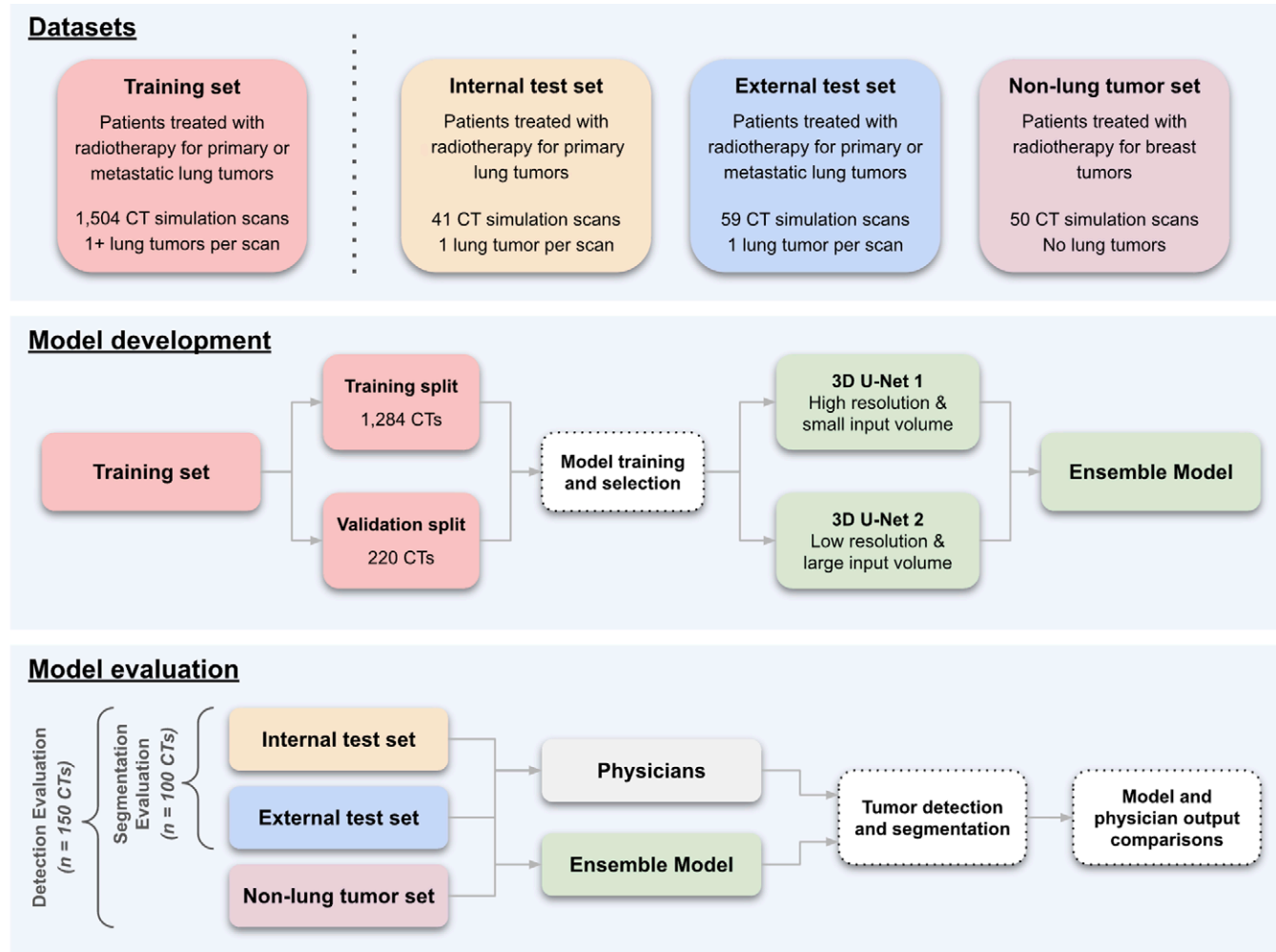


Figure 1: Diagram outlining the datasets and model development and evaluation framework. 3D = three-dimensional.

Model Development and Ensembling

Figure 1 provides an overview of the model development and evaluation framework. An ensemble consisting of two 3D U-Nets for lung tumor detection and segmentation was developed using the training set of 1504 CT scans. The two U-Nets had different properties: one was trained on high-resolution images with a relatively small input volume (original scans resampled to a voxel spacing of $1 \times 1 \times 2$ mm and a model input patch size of $192 \times 192 \times 96$ voxels), and the other was trained on low-resolution images with a relatively large input volume (original scans resampled to a voxel spacing of $2 \times 2 \times 3$ mm and a model input patch size of $192 \times 160 \times 80$ voxels). This ensemble approach was used to address GPU memory limitations and balance volume with resolution.

The model architecture closely follows the traditional U-Net structure. Batch and instance normalization were used for stability, and a leaky rectified linear unit, or ReLU, was used for including nonlinearities; the initial number of feature maps was set to 32, doubling on downsampling and halving on upsampling (Fig S2).

The models were trained for 100 epochs on NVIDIA Tesla V100 GPUs. The batch size was set to two, and Adam optimization was used, with an initial learning rate of 0.001; the learning rate was reduced by 90% every time the validation loss failed to improve for 10 consecutive epochs. Image augmentation was performed using the techniques listed in Table S1. For both models, the version with the lowest validation loss, defined as the sum of binary cross-entropy and Dice similarity coefficient (DSC) losses, was selected for further ensembling.

Model prediction was completed via a patch-based approach similar to the method described by Isensee et al (18). The models were then ensembled by averaging the per-voxel tumor prediction probabilities. A probability threshold of 0.5 was used for classifying a voxel as lung tumor.

A separate lung region isolation model (developed with methods similar to those used for the lung tumor models) was used to filter out predicted tumors found outside of the lungs. Predicted lesions smaller than 0.5 cm in diameter were also removed in the postprocessing phase, to decrease the number of false-positive findings due to noise. This minimum lesion size threshold was determined prior to model evaluation on the basis of previous studies describing the typical range of diameters for non-small cell lung cancer, in which fewer than 2% of lesions had a diameter less than 1 cm (19). The final postprocessed lung tumor prediction masks were used for subsequent model evaluation.

Model Evaluation

The lung tumor detection performance of the model was assessed in the combined test set of 150 CT scans (41 scans in the internal test set, 59 scans in the external test set, and 50 scans in the non-lung tumor test set). Specifically, model detection performance was evaluated by calculating sensitivity (eg, correct identification of a lung tumor on a scan, defined as detecting at least one true tumor voxel on a scan), specificity (eg, correct identification of scans without lung tumors), and the false-positive rate across all CT scans.

The segmentation performance of the model was evaluated on the 100 CT scans with lung tumors in the combined test set (41 scans in the internal test set and 59 scans in the external test set),

by comparing physician- and model-delineated lung tumor segmentations, using DSC. Specifically, the model segmentations for correctly identified lung tumors were compared with the clinical tumor segmentations used for radiation therapy planning. Given the known interobserver variability among physician-generated segmentations and, consequently, the lack of an objective reference standard label, additional radiation oncologists who were not involved in generating the clinical tumor segmentations were recruited to also contour each scan. Specifically, two additional radiation oncologists (M.F.G., 14 years of experience [study physician 1], and N.P., L.K.V., or A.C., 3, 8, and 9 years of experience, respectively [study physician 2]) segmented each scan, resulting in three physician segmentations per scan (one clinical segmentation and two research segmentations). The model segmentations were compared with all three physician segmentations to generate a model-physician DSC for each pairing.

To further benchmark model performance against physician performance, the interobserver variability of tumor segmentations was also evaluated between physicians. For both the internal and external test sets, DSCs were calculated for all physician segmentation pairings (eg, between the clinical segmentation and study physician 1 and 2 segmentations). These scores provide a reference for evaluating the segmentation accuracy of the model.

Code Availability

The code developed and used in this study is available at <https://github.com/mehrkashyap/lung-tumor-auto-detect-seg>.

Statistical Analysis

The main objectives of the statistical analyses were to evaluate the ability of the model to detect lung tumors on CT scans, assess the similarity of the model- and physician-generated tumor segmentations, determine the agreement between the model- and physician-determined tumor volumes, and compare the time required for the model and physicians to segment the tumors. Lesion-level metrics, including sensitivity, specificity, and false-positive rate, were calculated to evaluate model performance in detecting lung tumors. DSCs were calculated to assess the similarity of the model- and physician-derived lung tumor segmentations. The median DSCs and IQRs are reported. The concordance correlation coefficient was calculated to evaluate the agreement between model- and physician-determined tumor volumes, relative to the identity line (ie, $y = x$) (20). The Wilcoxon signed rank test was used to assess differences in volume between the model and physician segmentations. Bland-Altman plots were used to visually assess agreement between the model- and physician-determined lung tumor volumes. One-way analysis of variance was used to compare the mean time required for tumor segmentation among the model and physicians. All statistical analyses were performed by one author (M.K.) using Python 3.7.1 (Python Software Foundation) and SciPy 1.5.3. $P < .05$ was considered to indicate a statistically significant difference.

Results

Dataset Characteristics

A total of 1654 CT scans and clinical segmentations of 1928 lung tumors in 1445 patients from two affiliated medical centers were used to develop and evaluate the 3D U-Net ensemble model.

Table 1: Dataset and Lung Tumor Characteristics

Characteristic	Training Set	Internal Test Set	External Test Set	Non–Lung Tumor Set
No. of patients	1295	41	59	50
No. of CT scans	1504	41	59	50
CT Scanner	Siemens Biograph mCT or GE HealthCare Discovery	Siemens Biograph mCT or GE HealthCare Discovery	GE HealthCare LightSpeed	Siemens Biograph mCT
Median CT section resolution (mm)	0.98 × 0.98 × 1.5	0.98 × 0.98 × 1.0	1.27 × 1.27 × 2.5	0.98 × 0.98 × 2.0
No. of lung tumors	1828	41	59	0
Lung tumor type*				
Primary	827 (64)	41 (100)	50 (85)	
Metastatic	118 (9)	0 (0)	9 (15)	
Unknown	350 (27)	0 (0)	0 (0)	
Primary lung tumor characteristics				
AJCC stage†				
Stage 1	16 (6th ed) + 270 (7th ed)	21 (8th ed)	18 (8th ed)	
Stage 2	3 (6th ed) + 58 (7th ed)	7 (8th ed)	2 (8th ed)	
Stage 3	13 (6th ed) + 157 (7th ed)	8 (8th ed)	9 (8th ed)	
Stage 4‡	11 (6th ed) + 107 (7th ed)	5 (8th ed)	2 (8th ed)	
Unknown	192	0	19	
T stage†				
T1	14 (6th ed) + 250 (7th ed)	26 (8th ed)	NA	
T2	16 (6th ed) + 184 (7th ed)	4 (8th ed)	NA	
T3	3 (6th ed) + 88 (7th ed)	8 (8th ed)	NA	
T4	9 (6th ed) + 64 (7th ed)	3 (8th ed)	NA	
Unknown	199	0	50	
Lung tumor volume§				
<2 cm ³	389 (21)	8 (20)	5 (8)	
2–9 cm ³	571 (31)	13 (32)	7 (12)	
10–49 cm ³	469 (26)	13 (32)	13 (22)	
≥50 cm ³	399 (22)	7 (17)	34 (58)	

Note.—Except where indicated, data are numbers of tumors, with percentages in parentheses. AJCC = American Joint Committee on Cancer, ed = edition, NA = not available.

* Patient-level lung tumor type was determined using International Classification of Diseases, Ninth Revision, diagnosis codes for primary lung cancer (code 162) and lung metastasis (code 197). Lung tumor type for patients with both or neither of these codes was classified as *unknown*.

† Edition information in parentheses refers to AJCC staging manual edition.

‡ Patients labeled as having stage 4 primary lung cancer were distinct from those with metastatic lung tumors, as they lacked diagnostic codes for lung metastases, suggesting that radiation therapy was directed at the primary tumor rather than a metastatic lesion.

§ Equivalent spherical diameters: 2 cm³, 1.6 cm; 10 cm³, 2.7 cm; 50 cm³, 4.6 cm.

Dataset and patient characteristics are summarized in Tables 1 and 2, respectively.

Model Performance

Model lung tumor detection performance was evaluated on the 150 CT scans in the internal test set, external test set, and non–lung tumor set. The model achieved a sensitivity of 92% and a specificity of 82% for detecting lung tumors (Fig 2), and a mean of 0.78 false-positive tumors were detected per scan.

Model segmentation performance was assessed on the 100 CT scans with one lung tumor each from the internal and external test sets. The median DSC was 0.77 (IQR, 0.65–0.83) for all model-physician segmentation comparisons and 0.80 (IQR, 0.72–0.86) for all physician-physician comparisons (Table 3). Segmentation performance stratified by internal and external

test sets is shown in Table 4. Examples of model- and physician-generated lung tumor segmentations are provided in Figure 3.

Model performance was also assessed across different tumor volumes (Table S2). Detection of lung tumors across different volumes was consistently high (sensitivity ≥ 90%); however, segmentation of larger tumors was more variable (eg, DSC = 0.7 [IQR, 0.34–0.81] for lung tumors ≥ 50 cm³), driven largely by performance in the external test set.

The time required to segment lung tumors in the internal test set was shorter for the model than for the study physicians (mean, 76.6 seconds for the model vs 166.1 and 187.7 seconds for study physicians 1 and 2, respectively; one-way analysis of variance, $P < .001$ for both).

In an exploratory analysis, the model's ability to detect and segment multiple lung tumors on a scan was examined using a separate dataset (18 pretreatment CT scans from patients with

Table 2: Patient Characteristics

Characteristic	Training Set	Internal Test Set	External Test Set	Non–Lung Tumor Set
Median age (y)*	70.6 (62.1–78.4)	72.4 (65.6–78.4)	71.3 (63.5–79.1)	53.0 (47.1–63.6)
Sex				
Male	774 (60)	32 (78)	38 (64)	0 (0)
Female	509 (39)	9 (22)	21 (36)	50 (100)
Unknown	12 (1)	0 (0)	0 (0)	0 (0)

Note.—Except where indicated, data are numbers of patients, with percentages in parentheses.

* Data in parentheses are IQRs.

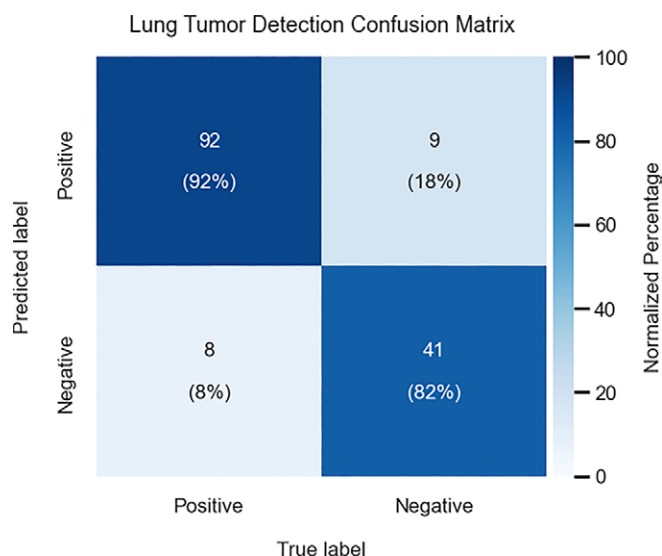


Figure 2: Per-scan lung tumor detection confusion matrix for the 150 CT scans in the combined test set. Percentages were determined by dividing absolute numbers by column sums.

metastatic non–small cell lung cancer undergoing immunotherapy; mean of 4.95 tumors per scan; median volume, 2.0 cm³) (Appendix S1, Table S3). In this dataset, the model achieved a sensitivity of 67% (60 of 89) in detecting tumors. The median DSCs for the model-generated segmentations compared with those obtained by study physicians 1 and 2 were 0.77 and 0.69, respectively (Table S4).

Tumor Volume Estimation

Agreement between model- and physician-measured tumor volumes was assessed for all correctly detected lesions in the internal and external test sets. The relationships between the model- and physician-measured tumor volumes for all model-physician and physician-physician pairings are shown in Figure 4. Bland-Altman plots are presented in Figures S3 and S4. The mean concordance correlation coefficient was 0.38 for the model-physician pairings and 0.93 for physician-physician pairings. The low agreement between model- and physician-measured tumor volumes was largely influenced by a subset of poor model segmentations, as the mean concordance correlation coefficient was 0.84 when the worst 15% of model segmentations based on DSC were excluded. A radiation oncologist reviewed these worst 15% segmentations to identify characteristics that led to poor model performance (Fig S5). Each of these tumors either abutted or had invaded the mediastinum or chest wall,

and most had adjacent postobstructive atelectasis and were very large. Overall, the model tended to underestimate tumor volume as defined by the clinical segmentation (Wilcoxon signed rank test, $P < .01$), with 61% of the tumors being larger than the model-predicted volume.

Discussion

We developed a deep learning–based three-dimensional U-Net multiresolution ensemble model to detect and segment lung tumors on CT scans. Our model was trained and evaluated on four datasets from two medical centers and showed robust performance in detecting and segmenting tumors.

To the best of our knowledge, our training dataset is the largest collection of CT scans and clinical tumor segmentations used to construct a lung tumor detection and segmentation model (17,21–26). This dataset was retrospectively collected from routine pretreatment CT simulation scans in patients undergoing radiation therapy and assembled through a scalable pipeline involving an automated data extraction process and a custom web application for efficient clinician review of segmentation quality. We believe that this pipeline can serve as a blueprint for other institutions to generate their own medical image segmentation datasets. As such, we have made our code available for others to use (<https://github.com/mehrkashyap/lung-tumor-auto-detect-seg>).

We believe that the large size of the training set and the heterogeneity of the data are key strengths of our approach, as these allowed the model to learn from the data of different patients and tumors, scanned with different devices and delineated with different styles of segmentation, as performed by the multiple radiation oncologists involved in creating the original clinical lung tumor segmentations for radiation therapy planning. Furthermore, using the original treatment segmentations ensured that the model learned from clinically acceptable contours used for patient care. Despite differences in scanning devices and practice patterns between the sites, the model demonstrated strong performance in both the internal and external test sets, suggesting that training on a large, heterogeneous dataset of clinical segmentations produced robust and generalizable results.

Our model has several other notable strengths that build upon those of existing works. A key advantage is the model's high sensitivity and specificity in detecting lung tumors without the need for human inputs such as seed point designation, a limitation of prior models (17). Moreover, our model achieved expert-level segmentation performance overall, though it performed worse

than experts in segmenting particularly large tumors. The ability to accurately segment lung tumors rapidly and automatically has important implications for quantifying tumor burden and assessing treatment responses in oncology, with automated processes potentially outperforming existing methods such as RECIST (Response Evaluation Criteria in Solid Tumors) (27,28), which are often labor-intensive, subjective, and limited by the use of unidimensional measurements (1–5).

We also highlight other innovations that set our model apart from prior models. First, the use of a 3D U-Net architecture distinguishes our work from approaches such as the one proposed by Primakov et al (25), which used a two-dimensional architecture.

Table 3: Comparison of Model- and Physician-generated Lung Tumor Segmentations for 100 Test Set Scans

Comparison	Median DSC for Detected Lesions
Model-physician comparisons	
Model vs clinical segmentation	0.78 (0.65–0.85)
Model vs study physician 1	0.76 (0.66–0.82)
Model vs study physician 2	0.77 (0.63–0.84)
Overall*	0.77 (0.65–0.83)
Physician-physician comparisons	
Study physician 1 vs clinical segmentation	0.80 (0.71–0.86)
Study physician 1 vs study physician 2	0.83 (0.74–0.87)
Study physician 2 vs clinical segmentation	0.81 (0.76–0.85)
Overall*	0.80 (0.72–0.86)

Note.—Data in parentheses are IQRs. For all model-physician and physician-physician segmentation pairings, DSCs were calculated across all CT scans. DSC = Dice similarity coefficient.

* The overall results are presented as the median of the mean DSC for each scan across all model-physician or physician-physician segmentation pairings.

Table 4: Comparison of Model- and Physician-generated Lung Tumor Segmentations, Stratified by Internal and External Test Sets

Comparison	Median DSC for Detected Lesions	
	Internal Test Set	External Test Set
Model-physician comparisons		
Model vs clinical segmentation	0.77 (0.67–0.85)	0.78 (0.62–0.85)
Model vs study physician 1	0.76 (0.68–0.80)	0.75 (0.60–0.84)
Model vs study physician 2	0.78 (0.73–0.84)	0.75 (0.55–0.83)
Overall*	0.77 (0.69–0.82)	0.75 (0.61–0.83)
Physician-physician comparisons		
Study physician 1 vs clinical segmentation	0.83 (0.72–0.86)	0.78 (0.70–0.85)
Study physician 1 vs study physician 2	0.85 (0.76–0.88)	0.80 (0.71–0.86)
Study physician 2 vs clinical segmentation	0.83 (0.76–0.87)	0.80 (0.76–0.85)
Overall*	0.80 (0.75–0.86)	0.79 (0.68–0.83)

Note.—Data in parentheses are IQRs. For all model-physician and physician-physician segmentation pairings, DSCs were calculated across all CT scans. DSC = Dice similarity coefficient.

* The overall results are presented as the median of the mean DSC for each scan across all model-physician or physician-physician segmentation pairings.

By capturing rich information beyond that available on individual CT sections, our 3D model is theoretically capable of identifying smaller lesions that two-dimensional models may be unable to distinguish from structures such as blood vessels and airways. Second, we employed an ensemble approach—incorporating a high-resolution model with smaller input volumes and a low-resolution model with larger input volumes—designed to both provide a balance between volumetric context and image resolution and address memory constraints that often limit the use of 3D models (18,29). Finally, our model was trained on a large dataset, a subset of which consisted of scans with multiple lung tumors. Consequently, in an exploratory dataset of patients with metastatic non–small cell lung cancer, our model demonstrated the potential for detecting and segmenting multiple lung tumors on scans. The ability to delineate multiple tumors is necessary for comprehensive assessment of tumor burden and for treatment monitoring; however, more comprehensive evaluations of the model in this context are needed, including studies with larger datasets and studies at external sites.

Despite its promising results, our study has notable limitations. One limitation is that we did not specifically address lung tumors that are difficult to contour on CT scans alone, such as those adjacent to soft tissue structures or associated with atelectasis, which could be more easily and accurately delineated on PET images. Since our dataset consisted of pretreatment CT simulation scans without corresponding PET images, the model's performance on these challenging cases remains to be assessed. Another limitation was the model's tendency to underestimate tumor volume and perform worse on very large tumors, particularly on a subset of cases in the external test set. This is notable, as larger tumors are the most labor-intensive to contour manually and are therefore the tumors for which radiation oncologists could benefit most from autosegmentation. These issues have important implications for tasks such as tumor burden quantification. Interestingly, since our training dataset included clinical segmentations completed by radiation oncologists, we expected the model to learn their contouring patterns. Given the potential clinical consequences

of missing tumor tissue during treatment, radiation oncologists may tend to overestimate tumor volumes to ensure adequate coverage. However, our model's underestimation of tumor volume suggests that other factors may have influenced its performance. It is possible that the 3D patch-based approach we used, which divides CT scans into overlapping volumes, hindered the ability of the model to segment larger tumors, since these may not completely fit into any one input patch. Additionally, large tumors present unique segmentation challenges, as they often invade the mediastinum or chest wall and are frequently accompanied by adjacent postobstructive atelectasis. As such, implementing our model in a physician-supervised workflow would be necessary, allowing clinicians to identify and discard incorrectly identified lesions and lower-quality segmentations. Finally, this was a retrospective study that, despite the rigor of our evaluation, could not capture the clinical

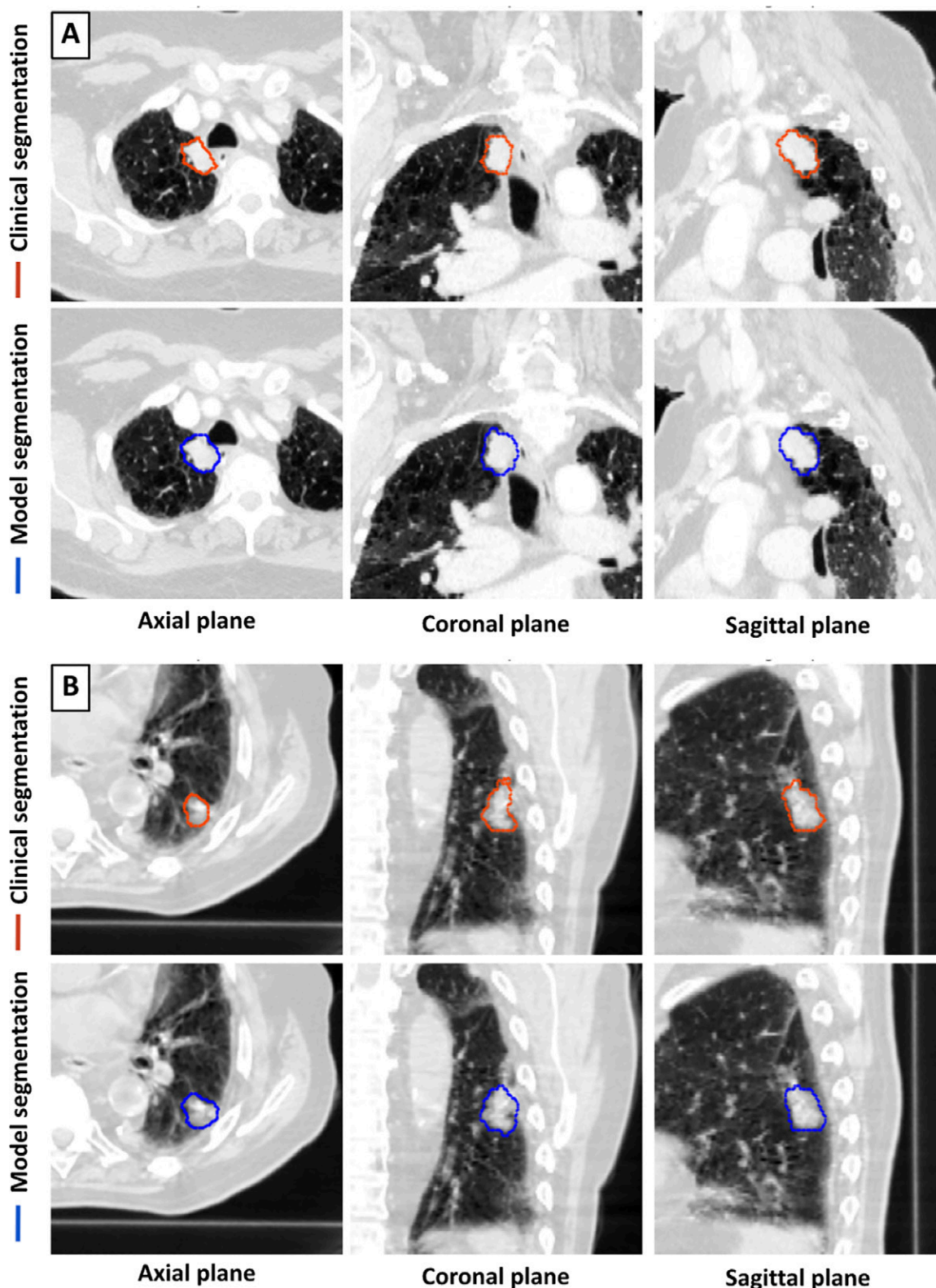


Figure 3: Example clinical (red) and model (blue) segmentations on CT simulation scans acquired prior to radiation therapy. Clinical segmentations represent the contours developed in the radiation therapy planning process. **(A)** Images in a 71-year-old female patient with non-small cell lung cancer from the internal test set. **(B)** Images in an 87-year-old male patient with non-small cell lung cancer from the external test set.

complexities that would be present in a real-world prospective deployment of the model.

Given that an important application of accurate automated tumor segmentation and volume measurement is assessing tumor response to therapy, future research should focus on applying our

model to estimating total lung tumor burden longitudinally and evaluating treatment response, and comparing it to existing methods such as RECIST. Prospective evaluation of the performance of our model is also necessary, as demonstrated by recent studies in the field (25). Furthermore, we believe it would be valuable to

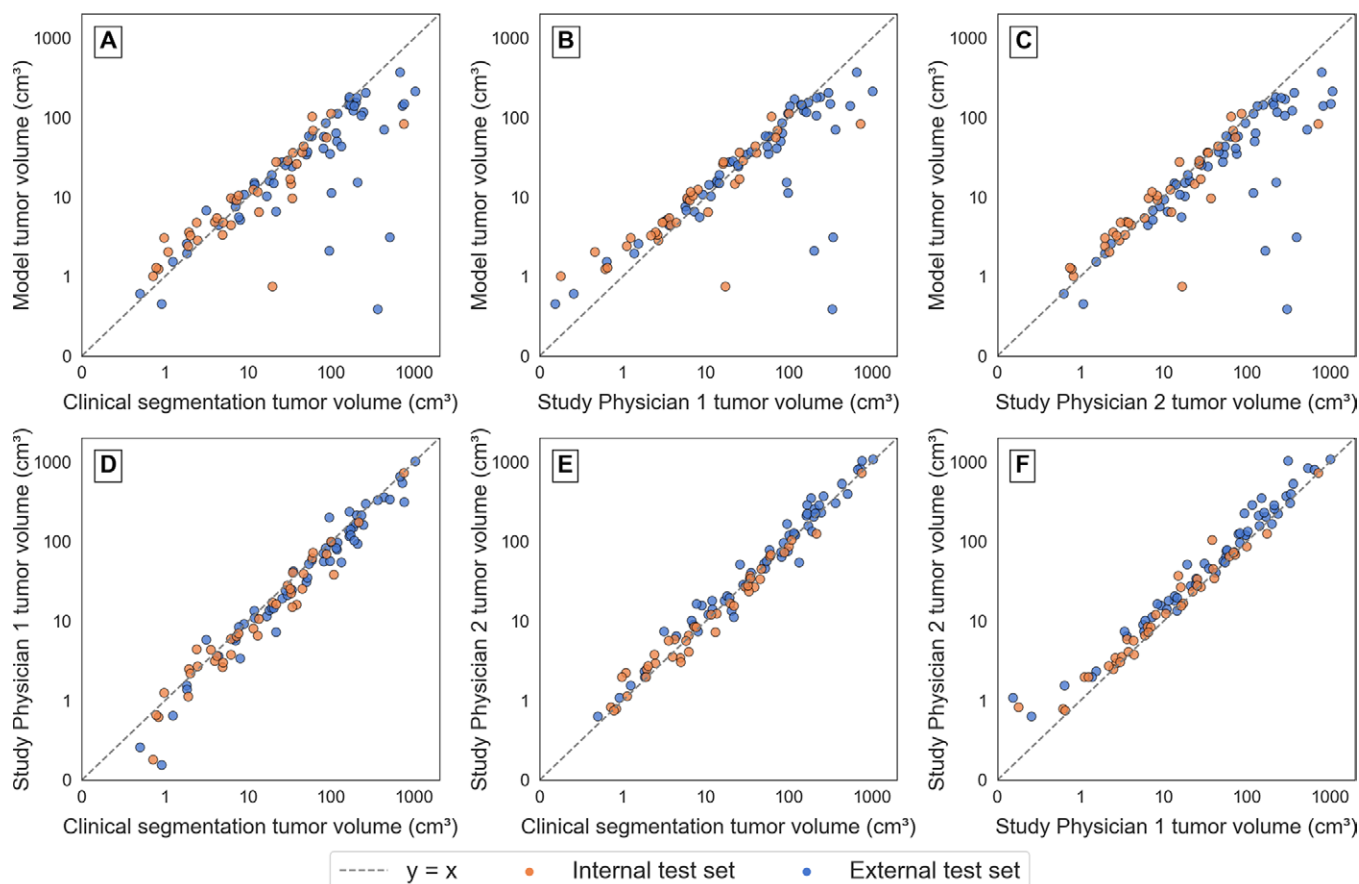


Figure 4: Comparison of model- and physician-determined lung tumor volumes. (A–C) Scatterplots show the relationships between model-determined tumor volumes and volumes determined by (A) clinical segmentation, (B) study physician 1, and (C) study physician 2. (D–F) Scatterplots compare physician-determined tumor volumes: (D) study physician 1 versus clinical segmentation, (E) study physician 2 versus clinical segmentation, and (F) study physician 2 versus study physician 1. The line represents perfect agreement ($y = x$) between the two volume measurements.

assess our model's ability to predict clinical outcomes such as survival on the basis of estimated tumor volume, particularly when combined with other prognostic models using diverse clinical data (30,31).

In conclusion, our study represents an important step toward automating lung tumor identification and segmentation. We demonstrated the functionality of a pipeline for extracting and reviewing routine pre–radiation treatment CT simulation scans and clinical segmentations to generate a large dataset for medical image segmentation. We developed a model using the resulting data, achieving performance comparable to that of expert clinicians, with generalizability to an external site with different processes from the internal site. This approach could have wide-ranging implications, including its incorporation in automated treatment planning, tumor burden quantification, treatment response assessment, and other radiomic applications.

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Received November 9, 2023; revision requested January 8, 2024; final revision received December 2; accepted December 30.

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Funding: Supported in part by the National Cancer Institute (grant number 5P30CA124435) and the National Center for Research Resources (Clinical and Translational Science Award number UL1 RR025744).

Acknowledgments: We would like to thank the members of the Stanford Medicine Department of Radiation Oncology for their collaboration and support throughout this study, and John Emmett Worth, BSHS, for assistance with recording cancer staging information.

Author contributions: Guarantors of integrity of entire study, M.K., X.W., Q.Z., K.B., L.K.V., M.E.G.; study concepts/study design or data acquisition or data analysis/interpretation, all authors; manuscript drafting or manuscript revision for important intellectual content, all authors; approval of final version of submitted manuscript, all authors; agrees to ensure any questions related to the work are appropriately resolved, all authors; literature research, M.K., X.W., R.L., M.E.G.; clinical studies, Q.Z., K.B., P.D., B.W.L., M.D., M.E.G.; experimental studies, M.K., X.W., N.P., S.Z.; statistical analysis, M.K., N.P., M.E.G.; and manuscript editing, M.K., X.W., M.H., C.H., A.C., L.K.V., S.Z., B.W.L., M.D., L.X., R.L., M.E.G.

Disclosures of conflicts of interest: M.K. No relevant relationships. X.W. No relevant relationships. N.P. No relevant relationships. M.H. No relevant relationships. Q.Z. No relevant relationships. C.H. No relevant relationships. K.B. No relevant relationships. A.C. No relevant relationships. L.K.V. Patents planned, issued, or

pending with Stanford University. **P.D.** No relevant relationships. **S.Z.** No relevant relationships. **B.W.L.** No relevant relationships. **M.D.** Grants or contracts from AstraZeneca; royalties or licenses from Roche and Foresight Diagnostics; consulting fees from AstraZeneca, Gritstone Bio, Illumina, and Regeneron; payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing, or educational events from Bristol Myers Squibb; support for attending meetings and/or travel from Foresight Diagnostics and Regeneron; multiple patent filings on biomarkers assigned to Stanford University; board member for Foresight Diagnostics; stock or stock options from CiberMed, Foresight Diagnostics, Gritstone Bio, and Perception Medicine; and invited faculty at Hokkaido University (Japan). **L.X.** No relevant relationships. **R.L.** Grants or contracts from the National Institutes of Health and Sanofi. **M.F.G.** Grants or contracts to institution from Varian Medical Systems and XRad Therapeutics and stock or stock options from Roche.

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